Toward a shorter proof of the sharp iid Berry–Esseen bound: fixed-target excess and a common noise measure

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Abstract

Subtracting the proposed Berry–Esseen bound before maximizing fixes the coefficient in the contact equation. We prove attainment for this excess problem, derive its zero-bias identity, and state exact conditions under which existing scalar estimates transfer to the new extremizer. Under those conditions, a first-failure argument needs no upper-excess entry bound. We also give a short proof of a sharp family of gap inequalities and retain a common positive measure when bounding conditional noise. These are analytic reductions. A complete finite base and uniform local exclusion remain to be established; this note does not prove the conjecture.

Scope of this note

We independently derive the proposed replacement of the extremal ratio by an additive excess over a fixed target. Attainment and closed-support contact hold for every positive target at each fixed index. The conserved identity has the coefficient $(n + H^2)a$, with H the raw sum threshold. This removes the actual score from the contact and remainder coefficients. It does not establish a new uniform local exclusion or automatically accept the old numerical contracts for a different maximizing law. All original source files and certificates are preserved.

1 Definitions and exact scope

Let $\mathcal{P}$ be the laws on $\mathbb{R}$ with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and $\rho = \mathbb{E}|X|^3 < \infty$. All sums below use independent copies of one marginal law. For $S_n = X_1 + \cdots + X_n$, set

$$F_{n,P}(H) = \mathbb{P}(S_n \le H), \qquad z = H/\sqrt{n}, \quad R_n(P, H) = \frac{\sqrt{n}}{\rho}\{F_{n,P}(H) - \Phi(z)\}.$$

The global Berry–Esseen constant at this index is the supremum of the absolute standardized discrepancies, divided by $\rho/\sqrt{n}$; call it C_n. Reflection and approximation by closed thresholds show that it is also $\sup_{P,H} R_n(P, H)$. For a fixed number $c > 0$ define

$$a(P, H; c) = \sqrt{n}\{F_{n,P}(H) - \Phi(z)\} - c\rho, \qquad A_n(c) = \sup_{P,H} a(P, H; c).$$

The excess a has the units of ρ times a score difference. It is *not* the existing manuscript's $a_n = C_n - C_{\mathrm{E}}$. At a maximizing pair of this new problem its actual positive score is

$$C_{\mathrm{act}} = c + a/\rho. \tag{1}$$

In general C_{act} need not equal C_n.

For conservation and defect positivity we later specialize to $c = C_{\mathrm{E}} = (3 + \sqrt{10})\phi(0)/6$. Attainment and contact themselves require only $c > 0$. No Fourier bound, numerical acceptance or uniform index limit is part of their proof.

2 Attainment and the equality boundary

Theorem 1. *For each fixed integer $n \geq 1$ and $c > 0$, the maximum $A_n(c)$ is attained by a pair (P, H) with a positive closed-CDF discrepancy. Moreover,*

$$A_n(c) > 0 \iff C_n > c, \qquad A_n(c) = 0 \iff C_n = c, \qquad A_n(c) < 0 \iff C_n < c.$$

Proof. Lyapunov gives $\rho \geq 1$, hence $A_n(c) \leq \sqrt{n} - c$. Symmetric Bernoulli, at $H = n$, gives the stronger lower bound

$$A_n(c) \geq \sqrt{n}\{1 - \Phi(\sqrt{n})\} - c > -c.$$

Write $A = A_n(c)$, and choose $a_j = a(P_j, H_j; c) \to A$. For all sufficiently large j, $a_j \geq A - 1$ and

$$c\rho_j \leq \sqrt{n} - a_j \leq \sqrt{n} - A + 1, \qquad \sqrt{n}\{F_{n,P_j}(H_j) - \Phi(H_j/\sqrt{n})\} = a_j + c\rho_j \geq a_j + c \geq \eta > 0,$$

where η can be any fixed number smaller than $A + c > 0$ once j is large enough. Unit variance implies tightness. A bound $\rho_j \leq M$ gives

$$\int_{|x|>K} x^2 \, dP_j \leq M/K, \qquad \int_{|x|>K} |x| \, dP_j \leq M/K^2.$$

Take subsequences $P_j \Rightarrow P$ and $\rho_j \to b$. Bounded continuous truncation and these tails preserve mean zero and variance one; lower semicontinuity gives $1 \leq \rho(P) \leq b < \infty$.

For $H_j < 0$, Chebyshev gives $F_{n,P_j}(H_j) \leq n/H_j^2$. For $H_j > 0$, the positive discrepancy is at most $1 - \Phi(H_j/\sqrt{n}) \leq n/H_j^2$. Its lower bound $\eta/\sqrt{n}$ thus bounds $|H_j|$. Take $H_j \to H$. Fixed n permits passage from marginal weak convergence to the sum law. For every $t > 0$, eventually $H_j \leq H + t$, so the closed-set inequality and then right continuity give

$$\limsup_j F_{n,P_j}(H_j) \leq F_{n,P}(H + t) \longrightarrow F_{n,P}(H).$$

Consequently

$$a(P, H; c) \geq A + c\{b - \rho(P)\} \geq A.$$

The definition of A forces equality, and $c > 0$ forces $\rho(P) = b$. Third-moment convergence is a *conclusion of maximality*; bounded third moments were not used to assert uniform integrability of third powers. Also $A + c\rho \geq A + c > 0$, proving positive discrepancy.

Reflection changes a negative closed discrepancy into a positive open discrepancy of the reflected law. Closed thresholds increasing to that open threshold approximate it; the penalty is unchanged. This proves the stated supremum identity for C_n without presuming continuity at an atom. If $C_n < c$, every excess is at most $\rho(C_n - c) \leq C_n - c < 0$. If $C_n > c$, a positive closed score above c gives positive excess. If $C_n = c$, choose scores $r_j \uparrow c$ with $r_j \geq c/2$. Their moments satisfy $\rho_j \leq 2\sqrt{n}/c$, and $\rho_j(r_j - c) \to 0$. Thus $A_n(c) \geq 0$, while all excesses are at most zero. These observations prove all three equivalences; no ratio-attainment lemma is needed for this last step. □

3 Atom-safe fixed-target contact

Fix a maximizing pair of Theorem 1, with $n \geq 2$. Set $F = F_{n,P}(H)$, $\varphi = \phi(z)$, $\beta = \mathbb{E}[X|X|]$ and $\gamma = \mathbb{E}X^3$.

Theorem 2. *The following function is nonpositive for every $y \in \mathbb{R}$ and vanishes on the entire closed support of P:*

$$I_c(y) = n^{3/2}\{F_{n-1,P}(H-y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2-1) + \tfrac{3}{2}c\rho(y^2-1) - c\{|y|^3 - \rho - 3y\beta\}. \tag{2}$$

That support is compact. Equivalently, for $h(y) = F_{n-1,P}(H-y)$, $h \le Q_c$ everywhere and $h = Q_c$ on the closed support, where

$$nQ_c(y) = nF + \frac{H\varphi + c\rho}{2\sqrt{n}} - \left(\sqrt{n}\varphi + \frac{3c\beta}{\sqrt{n}}\right) y - \frac{H\varphi + 3c\rho}{2\sqrt{n}} y^2 + \frac{c}{\sqrt{n}}|y|^3. \tag{3}$$

Proof. Contaminate the raw law by $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_y$. Its mean and variance are $m_\varepsilon = \varepsilon y$ and $v_\varepsilon = 1 + \varepsilon(y^2-1) - \varepsilon^2 y^2$. Let $r_\varepsilon = \mathbb{E}_{P_\varepsilon}|X - m_\varepsilon|^3$. The standardized law, evaluated at the standardized image of the *fixed raw event*, has excess

$$G_c(\varepsilon) = \sqrt{n}\left\{F_{n,P_\varepsilon}(H) - \Phi\left(\frac{H - nm_\varepsilon}{\sqrt{nv_\varepsilon}}\right)\right\} - c\frac{r_\varepsilon}{v_\varepsilon^{3/2}}.$$

For small $\varepsilon \ge 0$ this is admissible; maximality gives $G_c(\varepsilon) \le G_c(0) = a$. The raw CDF is a polynomial in ε and has derivative $n(h(y) - F)$. Dominated differentiation of $|x - \varepsilon y|^3$, bounded by $3|y|(|x| + |y|)^2$, gives

$$r_0' = |y|^3 - \rho - 3y\beta, \quad \left(\frac{H - nm_\varepsilon}{\sqrt{nv_\varepsilon}}\right)_0' = -\sqrt{n}\, y - \frac{H}{2\sqrt{n}}(y^2-1).$$

Differentiation now gives $G_c'(0+) = I_c(y) \le 0$. There is no division by ρ in this fixed-target derivative.

Integrating (2) against P gives zero. All its terms are integrable under the third-moment assumption. The map h is upper semicontinuous: at a jump the closed event chooses its larger value. Thus I_c is upper semicontinuous. If it were strictly negative at a closed-support point, a neighborhood of that point would have positive P-mass and a strictly negative upper bound for I_c, contradicting its zero integral. Hence it vanishes at every such point, including atoms and accumulation points. Because $c > 0$, a bound of the form $I_c(y) \le -c|y|^3 + M_2y^2 + M_1|y| + M_0$ excludes both sufficiently distant rays from the support. Rearrangement gives (3). □

Neither proof assumes $H \ge 0$. Reflecting a positive closed discrepancy again does not in general preserve that event. The actual orientation must be retained after choosing the maximizing pair.

4 Predecessor support: the extra term has a favorable sign

Theorem 3. *Suppose additionally $C_{n-1} \le c$. Put $v = n-1$, $s = \sqrt{n/v}$ and $e_n = n(s-1) - 1/2$. For every support point y,*

$$0 \le \mathcal{H}_n(H,y) + c\{-|y|^3 + \tfrac{3}{2}\rho y^2 + 3\beta y + \rho e_n\} - na, \tag{4}$$

where, with $F_v(x) = \Phi(x/\sqrt{v})$,

$$\mathcal{H}_n(H,y)/n^{3/2} = F_v(H-y) - F_{v+1}(H) + yF_{v+1}'(H) - \tfrac{1}{2}(y^2-1)F_{v+1}''(H).$$

Proof. For the same marginal, $F_{n-1,P}(H-y) \le F_v(H-y) + c\rho/\sqrt{v}$. Substitute this upper bound and $F = \Phi(H/\sqrt{n}) + (c\rho+a)/\sqrt{n}$ in the zero contact equality. The constant moment contribution is $c\rho\{n(s-1)-1/2\}$, and the new excess contributes exactly $-na$. The Gaussian derivatives give the displayed expression. This proves the claim for each sign of y and every real H. □

If $a \ge 0$, dropping $-na$ gives the old zero-loss support inequality with coefficient c. Therefore the elementary support bounds in current Section 4 apply after their hypotheses are restated as Theorem 2, $C_{n-1} \le c$, $a \ge 0$, and the same moment caps. At $c = C_{\mathrm{E}}$, the lower numerical multiplier $0.4095 < c$ remains valid only after its cubic coefficient is negative, as in the existing proof. The all-H fourth-order expansion, both ray signs, and frozen moment caps remain necessary. The structural moment bounds are universal bounds for each law's actual score, so (1) with $a \ge 0$ supplies their lower-score premise. This transfer does not re-prove those external structural inequalities or acquire the later conditional moment caps automatically.

5 Independent conservation calculation

This section needs only the explicit contact relations in Theorem 2, not the identity of the actual score with the contact coefficient. Set $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X \ge t\}}]$. Centering makes $\tau \ge 0$, and signwise Fubini gives $\int \tau = 1$ and

$$\int t\tau(t)\,dt = \gamma/2, \quad \int t^2\tau(t)\,dt = M_4/3, \quad \int |t|^3\tau(t)\,dt = M_5/4.$$

Here $M_4 = \mathbb{E}X^4$ and $M_5 = \mathbb{E}|X|^5$. These moments are available because compact support has already been proved. For each bounded complementary gap (u,v) of the closed support, τ is constant, say $\tau_{u,v}$. Define

$$J = \sum_{(u,v)} \tau_{u,v}(v-u)^2,$$

$$J_1 = \sum_{(u,v)} \tau_{u,v} \int_u^v (t-u)t\,dt, \qquad J_2 = \sum_{(u,v)} \tau_{u,v} \int_u^v (t-u)t|t|\,dt,$$

$$L = \sum_{(u,v)} \tau_{u,v} \int_{[u,v)} (t-u)\,d\nu(t), \qquad D = (3+\sqrt{10})\rho - \gamma - 3J.$$

For support in $[-R,R]$, $J \le 2R$, $|J_1| \le RJ/2$, $|J_2| \le R^2J/2$ and $0 \le L \le 2R$. Thus the sums converge absolutely. The gap/defect theorem supplies $D \ge 0$ when that sign is used below; the algebraic identity itself does not require that theorem.

Put $\mathcal{A} = \int \tau(Q_c - h)$. Contact implies

$$h(u) - h(v) = \nu([u,v)) = Q_c(u) - Q_c(v).$$

Integrating $h(t) = h(v) + \nu([t,v))$ across the gap gives

$$\int_u^v (Q_c - h) = \int_u^v (t-u)(-Q_c'(t))\,dt - \int_{[u,v)} (t-u)\,d\nu(t).$$

The half-open upper endpoint is essential. The slack is zero on the support and $\tau = 0$ outside its convex hull. Substituting Q_c' and summing gives the first evaluation

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{G_c}{n^{3/2}} - L, \quad G_c = \tfrac{3}{2}c\beta J + (H\varphi + 3c\rho)J_1 - 3cJ_2. \tag{5}$$

Let X^* have density τ, independently of S_{n-1}. Signwise Fubini gives $\mathbb{E}g'(X^*) = \mathbb{E}[Xg(X)]$. Conditional use of that identity and exchangeability show that $(S_{n-1}+X^*)/\sqrt{n}$ is the zero-bias law of $Z = S_n/\sqrt{n}$. Its CDF at z equals

$$T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{\{Z\le z\}}], \qquad \int \tau(t)h(t)\,dt = T.$$

The factor $Z^2 - zZ$ makes endpoint atoms harmless. Integrating the four coefficients of (3) gives

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3-M_4)/6 + cK_q}{n^{3/2}} - T, \quad K_q = \frac{M_5}{4} - \frac{\rho M_4}{2} - \frac{3\beta\gamma}{2} + \frac{\rho}{2}. \tag{6}$$

Multiplying the support contact equality by X and using exchangeability also gives, for $M = \mathbb{E}[Z\mathbf{1}_{\{Z\le z\}}]$,

$$M + \varphi = -\frac{\gamma z\varphi}{2\sqrt{n}} + \frac{cK_1}{n}, \quad K_1 = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma. \tag{7}$$

Define $q_z(x) = (x-z)|x-z|$, and let Z_0 be standard normal. Set

$$\varepsilon = \mathbb{E}q_z(Z) - \mathbb{E}q_z(Z_0) - \frac{2\gamma\varphi}{3\sqrt{n}}.$$

The pointwise hinge identity gives

$$T - \Phi(z) = -\varepsilon/2 - \frac{\gamma\varphi}{3\sqrt{n}} + z(M+\varphi) - z^2(F - \Phi(z)).$$

Substitute (7) and $\sqrt{n}(F-\Phi) = c\rho + a$ in (6). Direct collection gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2,\\ P_0 &= H^2(c\rho + a + \gamma\varphi/2) - H\{cK_1 + \varphi(M_4-3)/6\} + cK_q.\end{aligned}$$

Equating with (5) proves the following identity.

Theorem 4. *Let $U = J_1 + (M_4-3)/6$ and define*

$$\begin{aligned}\mathcal{R}_c(H) = c\{3\beta J/2 + 3\rho J_1 - 3J_2 - K_q\} + H(cK_1 + \varphi U)\\ - H^2(c\rho + \gamma\varphi/2) + E_n, \qquad E_n = -n^{3/2}\varepsilon/2.\end{aligned}$$

Then, for general $c > 0$,

$$(n+H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c(H). \tag{8}$$

In particular, at $c = C_{\mathrm{E}}$ and $w = H^2/(2n)$,

$$\boxed{\begin{aligned}(n+H^2)a + nc\rho(1-e^{-w})\\ + \frac{n\phi(0)e^{-w}}{6}D + n^{3/2}L = \mathcal{R}_c(H).\end{aligned}} \tag{9}$$

If $a \ge 0$ and the physical defect inequality $D \ge 0$ holds, all four terms on the left are nonnegative.

The na term comes from the ordinary discrepancy, and H^2a comes from the quadratic hinge. Replacing their sum by na in the exact identity is false; dropping H^2a gives a valid weaker inequality when $a \ge 0$. The formula uses the raw H; in standardized units the coefficient is $n(1+z^2)$.

5.1 What the Fourier representation still requires

The unchanged compensated-inversion lemma is the exact retained input at manuscript Section 5b, lines 143–200. It applies to a finite signed measure of zero mass and first moment with finite absolute second moment, for every real raw threshold. Applied to the sum law minus the Gaussian and cubic signed comparator, it gives

$$E_n = \frac{2\sqrt{n}}{\pi}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt, \qquad V_n(t) = f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6).$$

The signed comparator has mass one and first moment zero. Compact support already gives all required moments. No occurrence of the contact coefficient or the extremal score is present in this inversion step, so the existing proof applies without modification. This identity does not bound the integral. Complete low- and high-frequency envelopes, signed phase uniformity, domain coverage, moment gates and the infinite tail remain obligations. No nonlattice decay is available merely from the fixed-target reformulation.

6 Gaussian weight and phase price

Corollary 5. *Suppose $c = C_{\mathrm{E}}$, $a \ge 0$, $D \ge 0$, $n \ge N \ge 2$, and an independently established raw upper bound gives $\mathcal{R}_c \le r_0 < 2N/5$. Then*

$$(n + H^2)a + \frac{n\phi(0)}{6}D + n^{3/2}L \le \mathcal{R}_c.$$

Any later upper bound $\mathcal{R}_c \le r$ consequently gives $e^{-w} \ge 1 - 5r/(2n)$ and $nD \le 15000r/997$, using the same directed constants as the manuscript. If also $nD \le Nd_D$ and $0.4095\rho - d_D/15 > 0$, set

$$q_{\mathrm{extra}} = \frac{1 - 5r/(2N)}{2}(0.4095\rho - d_D/15) > 0 \quad \text{when } r < 2N/5.$$

A valid signed envelope $\mathcal{R}_c \le A(c,\rho) + F_{\operatorname{sgn} H}(c,\rho)|H| - Q(c,\rho)H^2$ implies, for $\kappa = 997/15000$,

$$(n + H^2)a + \kappa nD + n^{3/2}L \le A(c,\rho) + F_{\operatorname{sgn} H}(c,\rho)|H| - (Q + q_{\mathrm{extra}})H^2.$$

Proof. Equation (8) gives

$$\frac{\varphi}{6}(\gamma + 3J) = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n > c\rho - 2/5 > 0.$$

Therefore $c\rho - \phi(0)D/6 > 0$. Regrouping (9) leaves the additional nonnegative term $n(1 - e^{-w})(c\rho - \phi(0)D/6)$, which may be dropped. The density and defect bounds follow by retaining one nonnegative term at a time and using $c > 2/5$, $\phi(0) > 997/2500$. Finally $1 - e^{-w} \ge we^{-w}$ and $D \le d_D$ price that same term by $q_{\mathrm{extra}}H^2$. It is subtracted once, not counted twice. □

7 Transport of a zero-barrier scalar schedule

There is a useful distinction between the actual score C_{act}, the contact multiplier, and the deleted-sum constants. They coincide or are compared in the old derivation, but their roles can be separated. In this note the contact multiplier is always c. The predecessors remain global bounds for actual deleted convolutions. The actual score is given by (1); it is not set equal to c.

Theorem 6 (Conditional scalar-schedule transport)**.** *Set $c = C_{\mathrm{E}}$ and $N \ge 3$. Consider a finite, ordered, zero-barrier acquisition and exclusion schedule at every actual $n \ge N$. Suppose its proof has the following properties.*

1. *Its primitive moment, geometry, Fourier, count and concentration inequalities hold for every actual law in their stated domains. Its raw remainder estimates are universal scalar inequalities for the displayed $\mathcal{R}_t$, with a positive formal multiplier t in an explicitly covered interval containing c.*

2. *Its support steps use the zero-loss contact bound; its resource steps use the conserved identity, licensed weight removal, phase price, and nonnegativity of the excess. All moment/support domains are acquired in their stated order for the same law. Every denominator, sign, strict gap and uniform-index gate is checked.*

3. *A charged corner bound, if used, subtracts only a nonnegative charge $\chi(t)\rho$ with $\chi(c) = 0$, and its proved scalar envelope contains the point $t = c$. No positive lower excess barrier is used unless it has separately been established for this new witness.*

4. *The confinement proof uses the displayed derivative of the contact majorant with multiplier t and separate valid deleted-sum bounds. Its strict budget remains a strict upper envelope for $\mathcal{R}_c$ after the phase price.*

5. *Every positive-noise leaf strictly bounds the count right side with repair multiplier t on its domain, and its bound at $t = c$ is negative. The complete cover and the zero-noise strict binomial case are independently valid. On every positive-noise count leaf a separately established binomial bound gives $C_B \le c$.*

Then the same scalar bounds and unchanged numerical targets, if proved on those domains, exclude a maximizer of $A_n(c)$ with $a \ge 0$ under $C_{n-1}, C_{n-2} \le c$. An upper bound on C_{act} or on na/ρ is not an entry requirement for this transported schedule.

Proof. Initialize with Theorems 1–3 and the universal moment bounds. For a raw scalar inequality specialize its formal multiplier to $t = c$, without changing the actual marginal or any of its other parameters. Theorem 4 and Corollary 5 supply the required nonnegative resources; discarding $(n + H^2)a$ only weakens the resulting inequality. At a charged step the subtracted formal charge is zero. Thus the same endpoint enclosure dominates the specialized remainder. Support improves by the extra $-na$, which can be discarded. The explicit confinement derivative is the derivative of (3), with coefficient c, whereas the concentration estimates use C_{n-1}, C_{n-2} independently. A strict raw envelope remains strict after discarding any of these nonnegative terms. Induction over the finite ordered schedule establishes every successive enclosure for this law. Nothing is inferred from an unproved fixed point.

For the final count step, let Q be the difference from the binomial CDF and $f(u)$ its normal phase correction, as in the current count identity. Direct subtraction yields

$$\begin{aligned} a + \rho_B(c - C_B) = \sqrt{n}\pi_k\{Q/\pi_k - f(u)\} \\ + \sqrt{n}\left\{\Phi\left(\frac{\delta}{\sqrt{nv}}\right) - \Phi\left(\frac{\delta}{\sqrt{n(v+s)}}\right)\right\} + c(\rho_B - \rho). \end{aligned} \tag{10}$$

The right side is exactly the old comparison right side with formal multiplier $t = c$. Its strict negative upper bound contradicts $a \ge 0$ and $C_B \le c$. The nearest-count mass sign and whole-count curvature proofs therefore transport unchanged once their right-side inequalities are proved. At zero noise, maximizing over H for a fixed law still puts H at a binomial atom, because the penalty is independent of H; the separately assumed strict binomial comparison gives $a < 0$. □

Specific scalar bounds that meet the multiplier test. The continuous-cell atlas bounds the constant, linear and quadratic coefficients of $\mathcal{R}_t$ before using conservation. Its separate convexity in t and ρ is purely algebraic. Thus its four-corner bound remains valid at $t = c$ whenever the scalar interval contains c; it does not require C_{act} to lie in that interval. The two-interval bound in current Section 6 likewise still bounds $G(c)$: with $C_0 < c < \chi < C_1$,

$$G(c) \le \max\{G(C_0), G(\chi)\} \le \max\{G(C_0), G(\chi), G(C_1) - N(C_1 - \chi)\}.$$

This uses only the old low interval and its convexity, not an upper bound for C_{act}. The coherent six-corner bound also contains $c < 0.40974$, where its charge $n(t - 0.40974)_+$ is zero. The fixed-target calculation can eventually evaluate $G(c)$ directly, but the larger old bound remains sound if its primitive input bounds have been proved.

The signed cubic repair uses only $0 < c_* \le t \le c^*$ as a scalar multiplier condition. Setting $t = c \in (0.4, 0.4226)$ keeps both signs of the cubic coefficient valid; the negative case still uses its upper multiplier. No actual-score cap is needed for that version of the repair. The confinement multiplier interval $(0.4, 0.4253]$ also contains c; its two deleted-sum caps instead follow from the first-failure hypothesis.

Limits of this transfer. A scalar enclosure originally justified only at a ratio optimizer is not universal merely because it is stored as numbers. Its derivation must have the properties in Theorem 6. In particular, the complete low/high-frequency and floor-42 domains, the initial raw ceiling licensing weight removal, each strict budget, all acquired moment intervals, and the complete count cover are still inputs. Positive-barrier profiles with $K > 0$ or coefficient intervals excluding c are not covered. No assertion here accepts the old numerical predicates or all retained analytic candidates. This is a proved conditional transport, not an unconditional LOCAL42 theorem.

8 A replacement implication, not an established exclusion

Corollary 7. *Fix $N \ge 3$ and $c > 0$. Assume $C_j \le c$ for $j < N$. Assume independently that at every $n \ge N$ with $C_{n-1} \le c$ (and $C_{n-2} \le c$ if an argument uses that deletion), no maximizer of $A_n(c)$ has $a \ge 0$. Then $C_n < c$ for every $n \ge N$.*

Proof. Choose the first failure of the strict conclusion. All required predecessors are at most c. Theorem 1 provides the forbidden excess maximizer with $a \ge 0$. □

This new exclusion has no upper-score or upper-excess entry premise, so the one-step recurrence is unnecessary *for this conditional implication.* The old LOCAL42 assumption explicitly has both such premises and concerns a ratio maximizer. It cannot be substituted for the new assumption. Exclusion only of $a > 0$ yields a weak global bound, not the strict tail.

For a binomial comparator of score C_B and third moment ρ_B, direct subtraction gives $a = \rho_B(C_B - c) + \mathcal{E}_c$. The bound $\mathcal{E}_c \le \rho_B(c - C_B)$ yields only $a \le 0$. Strict exclusion requires a strict comparison or a separately proved strict margin. This equality boundary must not disappear during the reformulation.

9 Adversarial checks and remaining scope

The separate exact checker uses sparse Laurent polynomials over rational numbers to derive the zero-bias moments of Q_c, the truncated first moment, both slack expansions and the hinge elimination. A mutation using na leaves exactly $H^2 a$. Finite-law checks use symmetric and skew Bernoulli, its reflection, a three-point law with a zero atom, a four-point law, and three

symmetric tiny-far-atom laws. They check 108 zero-bias/hinge instances and 324 raw-mixture derivatives, at negative, zero, positive and colliding thresholds. Ninety-eight collisions detect the incorrect closed upper endpoint in the gap measure.

Two further controls identify analytic failure modes. For Bernoulli on $\{-1, 1\}$, the open-event function $y \mapsto \mathbb{P}(X < -y)$ at $y = 1$ has value zero and left limit $1/2$, so it is not upper semicontinuous. For integer $m > 1$, put mass $1/(2m^3)$ at each of $\pm m$, mass $(1 - 1/m)/2$ at each of ± 1, and the remaining mass $1/m - 1/m^3$ at zero. These standardized laws have $\rho_m = 2 - 1/m$, while the weak limit has $\rho = 1$; their third-moment tails at m equal one. This refutes the unjustified UI3 shortcut, not the attainment proof above.

Item	Mathematical status	Execution status
Fixed-index attainment, contact and support	Proved above; support uses its stated moment inputs	Exact finite-law checks pass
Conserved coefficient and sign gates	Exact identities and conditional consequences above	Independent rational algebra passes
Complete Fourier bounds and new local exclusion	Not established by this note	No native run or certificate acceptance

10 A short explanation of the gap defect

Let X be centered with variance one and finite $\rho = \mathbb{E}|X|^3$. Write $\gamma = \mathbb{E}X^3$ and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. The zero-bias density is $\tau(x) = \mathbb{E}[X\mathbf{1}_{X \geq x}]$. It is nonnegative, has integral one, and satisfies $2\int |x|\tau(x)\,dx = \rho$, by signwise Tonelli integration. On each bounded gap (a, b) of the closed support, it is constant; write this value as $\tau_{a,b}$ and set $J = \sum \tau_{a,b}(b - a)^2$.

Proposition 8 (The moment circle)**.** *For every such law,*

$$\gamma^2 + (2m)^2 \leq \rho^2, \qquad J \leq \rho + 2m.$$

Consequently, for every $t > 0$,

$$\gamma + tJ \leq (t + \sqrt{1 + t^2})\rho.$$

Equality in the last inequality holds exactly for a standardized two-point law whose positive mass is $p_t = (1 + t - \sqrt{1 + t^2})/2$.

Proof. Put $a_\pm = \mathbb{E}X_\pm^2$ and $b_\pm = \mathbb{E}X_\pm^3$. Then $a_+ + a_- = 1$, $\rho = b_+ + b_-$ and $\gamma = b_+ - b_-$. If $p = \mathbb{P}(X > 0)$ and $q = 1 - p$, Cauchy–Schwarz gives

$$a_\pm^2 \leq m b_\pm, \qquad m^2 \leq p a_+, \quad m^2 \leq q a_-.$$

Adding the last two inequalities after division by a_+ and a_- gives $m^2 \leq a_+ a_-$, and therefore $b_+ b_- \geq (a_+ a_-)^2/m^2 \geq m^2$. This proves the circle inequality.

For a gap on either side of zero, $(b - a)^2 \leq 2\int_a^b |x|\,dx$. If a central gap $(-A, B)$ exists, its weight is m, and its additional contribution is $2mAB$. The support condition gives $0 \leq \mathbb{E}[(X + A)(X - B)] = 1 - AB$. Summing over the countably many gaps gives $J \leq \rho + 2m$; in particular J is finite. Cauchy–Schwarz in two dimensions now gives $\gamma + 2tm \leq \sqrt{1 + t^2}\rho$.

Equality in the circle forces one nonzero atom on each sign class, and excludes an atom at zero. For the standardized two-point law, $J = \rho + 2m$, $m = \sqrt{pq}$, and $\gamma = (q - p)/\sqrt{pq}$. Equality in the last Cauchy bound requires $\gamma = 2m/t$, or $t(1 - 2p) = 2p(1 - p)$, whose unique admissible root is p_t. □

Thus, for $\sigma = \sqrt{10}$ and $p_* = (4 - \sqrt{10})/2$,

$$D = (3 + \sigma)\rho - \gamma - 3J = (\sigma\rho - \gamma - 6m) + 3(\rho + 2m - J)$$

is a sum of two nonnegative losses. The first measures departure from the extremal moment direction; the second measures inefficiency of the support gaps. Bounded support is unnecessary for this explanation. The quantitative endpoint inequality in the manuscript is still needed.

11 Keep the common measure when estimating conditional noise

Suppose now that a central gap $(-A, B)$ with $A, B > 0$ is known. Set $V = \mathbb{E}[X \mid \mathbf{1}_{X>0}]$, $W = X - V$, $\bar{A} = m/q$, $\bar{B} = m/p$, $\theta = \mathbb{E}W^2$, and $\Delta_A = \bar{A} - A$, $\Delta_B = \bar{B} - B$. Define positive measures

$$d\mu_+ = W^2 \mathbf{1}_{X>0}\, dP, \qquad d\mu_- = W^2 \mathbf{1}_{X<0}\, dP.$$

Write $\theta_\pm = \mu_\pm(\mathbb{R})$. These measures belong to the law, independently of the function whose expectation we wish to bound.

Proposition 9 (A weighted noise budget)**.** *Let*

$$\begin{aligned} B_0 &= \frac{(6 + 2\sigma)(1 - \theta)^{3/2}}{\sqrt{pq}}(p - p_*)^2, \\ E_0 &= 6m\{(A + B)(\Delta_A + \Delta_B) + \Delta_A\Delta_B\}, \\ k_+(x) &= (\sigma - 1)(x + 2\bar{B}) + 6B, \\ k_-(x) &= (\sigma + 1)(|x| + 2\bar{A}) + 6A. \end{aligned}$$

Then

$$D \geq B_0 + E_0 + \int k_+\, d\mu_+ + \int k_-\, d\mu_-.$$

Using $k_+ \geq 3(\sigma + 1)B$ and $k_- \geq 3(\sigma + 3)A$ gives the manuscript's $3(\sigma + 1)B\theta_+ + 3(\sigma + 3)A\theta_-$ bound.

Proof. Write $R_+ = \mathbb{E}X_+^2 - mB$ and $R_- = \mathbb{E}X_-^2 - mA$. The outer-gap calculation gives

$$D \geq (\sigma - 1)\mathbb{E}X_+^3 + (\sigma + 1)\mathbb{E}X_-^3 - 6mAB + 6BR_+ + 6AR_-.$$

Conditional centering gives the exact expansions

$$\mathbb{E}X_+^3 = m\bar{B}^2 + \int (x + 2\bar{B})\, d\mu_+, \quad \mathbb{E}X_-^3 = m\bar{A}^2 + \int (|x| + 2\bar{A})\, d\mu_-.$$

Also $R_+ = \mu_+(\mathbb{R}) + m\Delta_B$ and $R_- = \mu_-(\mathbb{R}) + m\Delta_A$. After substitution the deterministic terms equal $B_0 + E_0$: use $m^2 = (1 - \theta)pq$ and $(\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2$. The remaining terms are the displayed integrals. Since $x, \bar{B} \geq B$ on the positive class, $k_+ \geq 3(\sigma + 1)B$; the reflected bound is $k_- \geq 3(\sigma + 3)A$. □

For a twice continuously differentiable function h, set

$$r_h(v, x) = \int_0^1 (1 - u)h''(v + u(x - v))\, du.$$

Taylor's integral formula and conditional centering give

$$\mathbb{E}h(X) - \mathbb{E}h(V) = \int r_h(\bar{B}, x)\, d\mu_+ + \int r_h(-\bar{A}, x)\, d\mu_-.$$

This is valid when the terms are integrable, in particular on a bounded support for $h \in C^2$. At $x = v$ it includes the value $h''(v)/2$. Let $r = D - B_0 - E_0 \geq 0$. For any real responses $R_\pm$ and any finite nonnegative K satisfying $R_\pm(x) \leq K k_\pm(x)$ on their respective supports, integration proves

$$\int R_+ \, d\mu_+ + \int R_- \, d\mu_- \leq Kr.$$

One may take $K = \max\{0, \sup R_+/k_+, \sup R_-/k_-\}$ when finite. It is better to combine signed responses before taking this supremum: otherwise different terms spend the same variance at different, incompatible locations. This is a one-dimensional pointwise majorant problem; no strong duality or extremal-support theorem is required.

For a concrete comparison, consider a symmetric unit-variance law with masses $1/10$ at each of $-2, 2$ and $2/5$ at each of $-1/2, 1/2$. Here $A = B = 1/2$ and $\bar{A} = \bar{B} = 4/5$. For $h(x) = x^4$, the conditional response is $R(x) = x^2 + (8/5)x + 48/25$ in the magnitude coordinate. On $[1/2, 2]$ the positive-class ratio

$$\frac{R(x)}{(\sigma - 1)(x + 8/5) + 3}$$

is increasing: its derivative numerator is $(\sigma - 1)x^2 + 2dx + (16/25)(\sigma - 1) + 24/5 > 0$, where $d = (8/5)(\sigma - 1) + 3$. The negative-class denominator is larger. Thus the common bound is

$$K_{\text{weighted}} = \frac{228/25}{(18/5)(\sigma - 1) + 3} < \frac{228/25}{(21/10)(\sigma - 1) + 3} = K_{\text{minimum}}.$$

The right-hand expression is what results from replacing the weights by their actual minima on this interval before estimating the fourth moment. Using the still coarser coefficient $(3/2)(\sigma+1)$ from the scalar endpoint bound would give a larger constant again. The strict improvement is analytic, not an optimization sample. This example illustrates the gain; it is not a replacement of a critical Berry–Esseen certificate cell.

This argument does not linearize the convolution. If $f = \mathbb{E}e^{itX}$, $g = \mathbb{E}e^{itV}$ and $\delta = f - g$, it gives $|\delta| \leq t^2\theta/2$. For an integer $n \geq 2$, the exact remaining term is

$$f^n - g^n - ng^{n-1}\delta = n(n-1)\delta^2 \int_0^1 (1-u)(g + u\delta)^{n-2} \, du.$$

The segment $g + u\delta$ consists of characteristic functions of mixture laws, so its modulus is at most one. Thus

$$|f^n - g^n - ng^{n-1}\delta| \leq n(n-1)t^4\theta^2/8.$$

This bound is useful on a finite frequency interval. It does not supply the high-frequency decay or the cancellations required by a singular Fourier inversion kernel. Those remain separate obligations.

12 Spend the binomial margin instead of proving domination

Let $c = C_{\mathrm{E}}$, $p \in (0,1)$, $q = 1 - p$, $v = pq$ and $\rho_B = (p^2 + q^2)/\sqrt{v}$. At a legal binomial count k, put $z_B = (k - np)/\sqrt{nv}$ and $C_B = \sqrt{n}\{F_B(k) - \Phi(z_B)\}/\rho_B$. For any standardized law with this sign probability, at any threshold H, define

$$\mathcal{E}_c = \sqrt{n}\{F_{n,P}(H) - F_B(k) - \Phi(H/\sqrt{n}) + \Phi(z_B)\} + c(\rho_B - \rho).$$

Direct subtraction gives

$$a = \sqrt{n}\{F_{n,P}(H) - \Phi(H/\sqrt{n})\} - c\rho = \rho_B(C_B - c) + \mathcal{E}_c.$$

It is therefore enough to prove $\mathcal{E}_c < \rho_B(c - C_B)$ to exclude $a \geq 0$. Replacing $<$ by $\leq$ excludes positive excess only; it does not exclude equality. In particular, the strict tail conclusion needs a strict step.

For $p \in [1/3, 2/3]$, use the stronger binomial envelope

$$c(p) = \frac{\phi(0)(3 + |q - p|)}{6(p^2 + q^2)}.$$

The published binomial estimate supplies $C_B \leq c(p)$ on this interval [1, 2]. The algebraic margin is exact: with $r_p = \min(p, q)$,

$$c - c(p) = \frac{2c(r_p - p_*)^2}{p^2 + q^2}, \qquad \rho_B(c - c(p)) = \frac{2c(r_p - p_*)^2}{\sqrt{pq}}.$$

To check it, expand $(3 + \sqrt{10})(p^2 + q^2) - (3 + q - p) = 2(3 + \sqrt{10})(p - p_*)^2$ for $p \leq 1/2$ and reflect. Hence $\mathcal{E}_c < 2c(r_p - p_*)^2/\sqrt{pq}$ suffices for strict exclusion on this interval. If a strict version of the binomial envelope is used, a non-strict bound on $\mathcal{E}_c$ suffices instead.

The allowance vanishes at p_*, so a uniform argument near the extremal law is still necessary. Outside $[1/3, 2/3]$, this particular envelope cannot be used. No count panel is removed merely by displaying the factorization. A complete proposal must bound the actual comparison error, including phase, noise, endpoint atoms and normal variance correction, against the available margin.

13 Two tempting shortcuts, and what can replace them

13.1 A law need not be dominated by its own sign skeleton

The same sign probability is a natural way to choose a Bernoulli comparator, but it does not give a universal ordering of Berry–Esseen scores. Here $R_n(P) = \sup_H |\sqrt{n}\{F_{n,P}(H) - \Phi(H/\sqrt{n})\}|/\rho(P)$. For a fixed integer $n \geq 1$, let P_ϵ put mass ϵ at each of $\pm(2\epsilon)^{-1/2}$ and mass $1 - 2\epsilon$ at zero. It is centered, has variance one, and has third absolute moment $(2\epsilon)^{-1/2}$. The sum is symmetric and its zero atom tends to one as $\epsilon \downarrow 0$. Its Kolmogorov discrepancy consequently tends to $1/2$: symmetry bounds it above by $1/2$, while the closed edge at zero bounds it below by half the mass at zero.

Its sign skeleton has Bernoulli parameter $p = \epsilon$. The first standardized binomial atom tends to zero from below and has mass $(1 - \epsilon)^n \to 1$. At and on either side of that atom, monotonicity of the Gaussian CDF shows that the skeleton discrepancy also tends to $1/2$. Its third absolute moment is asymptotic to $\epsilon^{-1/2}$. Therefore, for every fixed n,

$$\frac{R_n(P_\epsilon)}{R_n(\text{sign skeleton})} \longrightarrow \sqrt{2} > 1.$$

Both scores tend to zero. This is an obstruction to universal skeleton domination, not to the sharp Berry–Esseen bound. A local comparison restricted to competitive laws remains possible.

There are small exact examples. At $n = 2$, the raw law on $-1, 0, 1$ with weights $1/20, 9/10, 1/20$ has closed event probability $363/400$ at zero. After standardization its score there is $163/(400\sqrt{5}) > 0.182239540166$. Every edge of the corresponding binomial law with $p = 1/20$ has score less than 0.180402292891. The accompanying rational checker encloses the Gaussian values and checks all edges. A version with no centered zero atom, and an already standardized rational example at $n = 42$, are given in the audit. The local hypotheses therefore have real work to do.

13.2 A continuum contact test has only finitely many candidates

For a fixed finite law and fixed calibrated parameters, one can replace a grid search over the contamination point by an exact finite test. Between the jumps of $F_{n-1,P}(H-y)$ and the point zero, the contact function, after multiplication by its positive normalization, is

$$d + ay + by^2 - c|y|^3, \qquad c > 0,$$

where d is constant on that interval. Its derivative is $a + 2by - 3cy^2$ on the positive half-line and $a + 2by + 3cy^2$ on the negative half-line. There are at most four candidate stationary points. At a jump, the closed-event value is the larger left value, and both tails tend to minus infinity. Hence the supremum occurs among zero, the closed jump values and the admissible stationary points. For a Bernoulli law the deleted sum has n atoms, giving at most $n + 5$ candidates in total.

This is a genuine reduction of the contact calculation. It still requires exact event ordering and rigorous enclosures for the candidate parameters, roots and Gaussian values. Even a successful continuum check proves stationarity only. The convolution objective is nonlinear in the common marginal, so a global extremal theorem needs an additional argument.

13.3 Why a common index parameter is not enough

It is sound to formulate first failure at an arbitrary index floor. It does not follow that a numerical exclusion valid from 42 works from 25. For example, the proposed simplification of the deletion factors has the wrong direction:

$$\left(\frac{N}{N-1}\right)^3 - \frac{N}{N-2} = \frac{N(N^2-3N+1)}{(N-1)^3(N-2)} > 0 \quad (N \geq 3).$$

At $N = 42$, the proposed substitute factor $103/100$ exceeds $\sqrt{42/40}$ but is smaller than $(42/41)^{3/2}$. Similarly, replacing a sharper finite-index Gaussian remainder $1/15$ by the universal $1/2$ changes the retained fourth-index support endpoint from $-51557/9765625$ to $25081283/58593750 > 0$. Neither replacement preserves the proof. A unified statement must retain the stronger numerical condition or prove a new estimate.

The public Lean developments offer useful models for organizing hypotheses. Their endpoints do not close the present finite base or LOCAL42 condition: one concerns eventual sharpness, one proves the larger constant 0.423, and the Stein-method library supplies different, non-sharp bounds. The accompanying source audit records pinned revisions and the exact interfaces inspected. No Lean build or formal verification of the new results is claimed.

14 The compensated inversion identity used above

For completeness, the Fourier identity in the fixed-target calculation has the same short proof as in the manuscript. Let μ be a finite signed measure with zero mass and first moment, and finite absolute second moment. Then, for every real H,

$$\int (x-H)|x-H|\, d\mu(x) = -\frac{4}{\pi}\int_0^\infty \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

The right side is absolutely convergent. Indeed, scaling and one integration by parts give

$$\int_0^\infty \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|.$$

To check the constant, use $t^{-2} = \int_0^\infty s e^{-st}\, ds$ and Tonelli to obtain

$$\int_0^\infty \frac{1-\cos t}{t^2}\, dt = \int_0^\infty \left(1 - \frac{s^2}{s^2+1}\right) ds = \frac{\pi}{2}.$$

For fixed y, the compensated integrand has its sign, so its absolute integral is $\pi y^2/4$. Hence Fubini against $|\mu|$ is justified by $\int (x-H)^2\, d|\mu|(x) < \infty$. The linear term vanishes by the two zero moments, proving the identity and absolute convergence.

Applied to the sum law minus its Gaussian and cubic signed comparator, this proves the inversion step in the conservation calculation. The characteristic function of the latter comparator is $e^{-nt^2/2}(1 - in\gamma t^3/6)$; its mass and first moment agree with those of the sum. This identity does not provide any numerical upper bound on its integral. Such a bound still needs the stated low-frequency and high-frequency estimates.

Use of AI. AI systems were used substantially in developing, checking and writing this note. The external suggestions were treated as proof candidates: the accompanying audits record accepted arguments, counterexamples and remaining assumptions. Independent AI review is not human peer review or formal verification.

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